

Armin Lotfy

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Professional Summary

Ph.D. Electrical Engineer and AI/Control specialist building **Python-based, data-driven and adaptive control** for **partially observable** and safety-critical systems. Expert in **RL/DL (PPO, SAC, TD3), EKF/UKF, MPC, and stochastic optimization (GA/PSO)**, with production-grade implementations in **TensorFlow, RLlib, and MATLAB/Simulink**. Design **POMDP** formulations, learnable controllers, and **digital-twin** simulations; apply **time-series forecasting, and multivariate statistics** to close the loop from **ingest, model, optimize, deploy**. Comfortable with **GPU-aware training, Optuna** library, reproducible pipelines, and **Git/GitHub CI**. Familiar with **agentic AI** (LLMs, tool use, multi-agent orchestration) and workflow automation (e.g., **n8n**) to accelerate research and ops.

Professional Experience

09/2020 – 09/2025

Ottawa, Ontario

Research Assistant

Carleton University

- Formulated multi-objective, partially observable control problems using POMDP frameworks with uncertainty and state-estimation considerations.
- Modeled electric vehicles, batteries, and hybrid energy-storage systems in Python, including signal preprocessing and dynamic system identification.
- Implemented plant–controller architectures in MATLAB/Simulink (Simscape) and Python; built digital-twin testbeds and scenario generators for system testing.
- Designed and deployed PPO, SAC, and TD3 actor–critic policies (feed-forward & LSTM) with reward shaping, safety filters, and domain randomization; trained models with Ray/RLlib and TensorFlow/Keras, and tuned with Optuna.
- Developed predictive state-estimation models using data-driven methods, filtering techniques, and time-series analysis for improved accuracy and robustness.
- Applied MPC, stochastic/robust optimization, and metaheuristic algorithms (GA/PSO) for control and energy-management optimization.
- **Thesis: “Optimization of Energy Management and Battery Cell Balancing in Battery Electric Vehicles Using Reinforcement Learning Techniques.”**

09/2022 – 04/2024

Ottawa, Ontario

Teaching Assistant

Carleton University

- Mentored 600+ students across Mechatronics & Circuits (ECOR 1043/1044, Python) and Switching Circuits & Digital Design (ELEC 2607, VHDL).
- Led labs and project coaching, guiding implementations on Raspberry Pi and Xilinx FPGA platforms.

09/2022 – Present
Ottawa, Ontario

Technical Support Clerk and Exam Site Facilitator

Carleton University

- **Providing** frontline technical support for students/faculty—**diagnosing and resolving** HW/SW issues, **mentoring** exam-tool usage, **documenting** fixes; **triaging** requests and **escalating** complex cases for timely, clear resolution.
- **Coordinating** multi-site exam logistics (proctors/instructors, room setup, compliance/accommodations); **managing** real-time incidents in high-volume settings, **minimizing** disruption and **keeping** sessions on schedule.

Skills

Technical Skills

Intelligent Control & Modeling

- Predictive modeling, time-series forecasting
- Data-driven and adaptive control system design

Reinforcement Learning & Deep Learning

- RL: TD3, MATD3, DDPG, PPO, MAPPO, SAC
- DL: CNN, RNN, LSTM

Programming & Tools

- MATLAB, Simulink, Python (NumPy, SciPy, Pandas, Matplotlib)
- TensorFlow, RLlib, Stable-Baselines3, SQLite3
- Git, Docker, VS Code, Linux Bash, LaTeX, VHDL

Large Language Models

- LLMs, RAG, fine-tuning; n8n, Ollama

Projects

RL-Based Energy Management System for BEVs — IRES Lab

- **Optimized power distribution and efficiency** under variable loads by developing a MATLAB/Simulink energy-management system powered by multi-agent reinforcement learning.
- **Ensured stable, deployable control** by training two TD3 agents—one for power-flow regulation and one for SOC balancing—achieving real-time-viable performance.
- **Matched DP-level accuracy**, achieving a final SOC within **0.3%** of a dynamic-programming baseline **without prior system knowledge or a full model**.

CTDE-Based Multi-Agent BMS for SOC/SOH Equalization — IRES Lab

- **Optimized current allocation while preserving battery health** by applying PPO in MATLAB/Simulink to a multi-objective control problem, using weighted scalarization and constraint penalties, and validating performance via closed-loop simulations.
- **Coordinated safe and efficient behavior under uncertainty** by formulating a shared-reward POMDP with belief-state tracking to handle sensor noise, enabling joint optimization of efficiency, safety, and degradation.
- **Reduced cell-to-cell SOC mismatch from 7% to 0.51%** without degrading overall driving performance.
- **Improved battery capacity degradation by ~39%** compared with a single-agent baseline method.

POMDP Controller for BEVs with Battery–Supercapacitor HESS (Ray/RLlib) — IRES Lab

- **Enabled uncertainty-aware control** by formulating the problem as a POMDP and training PPO with history stacks and predictive features to operate reliably under partial observability.

- **Reduced battery degradation by about 2%** while maintaining system performance and operational stability.
- **Developed real-time SOC/SOH forecasting and estimation** using CNN-LSTM models and an EKF for online inference; implemented in Python (RLlib, TensorFlow 2, Gymnasium) and validated in closed-loop using RMSE/MAE for deployment readiness.

Enhancing Power Sharing in Microgrids - IRES Lab

- Developed a PPO-based microgrid controller that optimally shares power among distributed resources under dynamic load and renewable fluctuations.
- Modeled the problem as an MDP in MATLAB/Simulink with well-defined states, actions, constraints, and reward shaping for efficiency and voltage/frequency stability.
- Achieved **27%** Integral of Absolute Error (IAE) reduction in islanded mode and **36%** in grid-connected mode compared to the conventional controller.

Education

2020 – 2025	Ph.D. in Electrical Engineering <i>Carleton University</i>
2016 – 2019	M.Sc. in Electronic Digital Engineering <i>IUST</i>
2010 – 2016	B.Sc. in Electronic Engineering <i>BNUT</i>

Publications

2025	Centralized Multi-Agent SOC Control for Battery Health Using Proximal Policy Optimization in EVs  <i>IEEE Transactions on Vehicular Technology</i> Lotfy, A., Alzayed, M., Chaoui, H., & Boulon, L.
2024	Enhancing Energy Management Strategy for Battery Electric Vehicles: Incorporating Cell Balancing and Multi-Agent Twin Delayed Deep Deterministic Policy Gradient Architecture  <i>IEEE Transactions on Vehicular Technology</i> Lotfy, A., Chaoui, H., Kandidayeni, M., & Boulon, L.
2025	Hierarchic Multi-Agent Energy Management for Extended Driving Range through Battery Cell Balancing  <i>34th IEEE International Symposium on Industrial Electronics (ISIE 2025)</i> Lotfy, A., Chaoui, H., Kandidayeni, M., Boulon, L., Khaneghah, M. Z., & Hasheminasab, S.
2025	Cutting-Edge EMS Technologies for EVs and HEVs: Recent Developments and Future Directions  <i>IEEE Access</i> Y. Farajpour, A. Lotfy , H. Chaoui and S. Kelouwani
2025	A Proximal Policy Optimization-Based Controller for Enhanced Power Sharing in Microgrids  <i>IEEE Access</i> S. Hasheminasab, A. Lotfy, M. Alzayed and H. Chaoui,

Courses

2020 Ottawa, Canada	Digital Signal Processing (A+) <i>University of Ottawa</i>
2021 Ottawa, Canada	Cryptographic Engineering and Side-Channel Analysis (A+) <i>Carleton University</i>
2021 Ottawa, Canada	Topics in Systems & Controls <i>Carleton University</i>

Awards

2025	Senate Medal nominee for Outstanding Graduate Work (Doctoral Level) <i>Carleton University</i> Senate Medal Nominee for Outstanding Graduate Work (Doctoral Level), recognizing exceptional doctoral research and academic achievement; nomination endorsed by the department and submitted to the university's Senate for final consideration.
2025	CMC Electronics Inc. Scholarship <i>Carleton University</i> CMC Electronics Scholarship recognizing academic excellence in Electrical & Computer Engineering; conferred by the Faculty of Engineering Scholarships Committee based on departmental recommendation.

Non-Profit Organizations

2025	Member of the Technical Program Committee, IEEE 102nd Vehicular Technology Conference (VTC2025) Served as a Technical Program Committee (TPC) Member for the IEEE 102nd VTC2025
2025	IEEE Vehicle Power and Propulsion Conference (VPPC) 2025 Reviewer Contributed as a Technical Reviewer for IEEE VPPC 2025, an internationally recognized conference in electric mobility and powertrain systems
2022	IEEE Vehicle Power and Propulsion Conference (VPPC) 2022 Reviewer Served as a Technical Reviewer for the IEEE Vehicle Power and Propulsion Conference (VPPC) 2022

References

Professional references and referee contact information available upon request.